AN ADMISSIBLE BLOWUP OVER AN OPEN SUBSCHEME EXTENDS TO THE WHOLE SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Gluing quasi-coherent sheaves from an open cover. Sheaves, Section sheaves-section-glueing-sheaves, used at proof lines 9140-9143. Produces the ideal I' on U cup V.
- Extension of a finite type ideal sheaf from an open subscheme. Properties, Lemma properties-lemma-extend, used at proof lines 9143-9146. Produces the finite type ideal J on X.

Source reference. Original lemma 01PF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2228–2228. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 00AK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 392–392. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $U, V \subset X$ be quasi-compact open subschemes. Let $b : V' \to V$ be a $U \cap V$-admissible blowup. Then there exists a U-admissible blowup $X' \to X$ whose restriction to V is V'.*

Proof. Let $\mathcal{I} \subset \mathcal{O}_V$ be the finite type quasi-coherent sheaf of ideals such that $V(\mathcal{I})$ is disjoint from $U \cap V$ and such that V' is isomorphic to the blowup of V in $\mathcal{I}$. Let $\mathcal{I}' \subset \mathcal{O}_{U \cup V}$ be the quasi-coherent sheaf of ideals whose restriction to U is $\mathcal{O}_U$ and whose restriction to V is $\mathcal{I}$ (see Sheaves, Section 00AK). By Properties, Lemma 01PF there exists a finite type quasi-coherent sheaf of ideals $\mathcal{J} \subset \mathcal{O}_X$ whose restriction to $U \cup V$ is $\mathcal{I}'$. The lemma follows. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.